

A post-quantum encryption scheme based on linearized Reed-Solomon codes

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- Post-quantum encryption scheme design tools
 - Errors correcting codes
 - Lattices
 - Isogeny
 - Hash functions
 - Multivariate systems

Introduction

- Post-quantum encryption scheme design tools
 - Errors correcting codes
 - Lattices
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 - Hash functions
 - Multivariate systems
- Code based cryptography
 - McEliece's scheme (1978)

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 - McEliece's scheme (1978)
 - HQC (NIST standard)

Outline

- 1 Generalities
- 2 Linearized Reed-Solomon codes and their distinguishability
- 3 New encryption scheme
- 4 Security

1 Generalities

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Linear codes

Let $\mathbb{F}_{q^m}$ a finite field and $k \leq n \in \mathbb{N}^*$.

- A **linear code** $\mathcal{C} \subset \mathbb{F}_{q^m}^n$.

$$\mathcal{C} = \{\mathbf{mG} \mid \mathbf{m} \in \mathbb{F}_{q^m}^k\} = \{\mathbf{c} \in \mathbb{F}_{q^m}^n \mid \mathbf{Hc}^T = 0\}$$

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- **Dual** of $\mathcal{C}$:

$$\mathcal{C}^\perp = \{\mathbf{x} \in \mathbb{F}_{q^m}^n \mid \forall \mathbf{c} \in \mathcal{C}, \langle \mathbf{c}, \mathbf{x} \rangle = 0\}$$

$$\text{where } \langle \mathbf{x}, \mathbf{y} \rangle := \sum_{i=1}^n x_i y_i, \forall \mathbf{x}, \mathbf{y} \in \mathbb{F}_{q^m}^n$$

Some metrics

Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{F}_{q^m}^n$.

- Rank weight of $\mathbf{a}$ over $\mathbb{F}_q$: $w_R(\mathbf{a}) := \dim \left(\langle a_1, \dots, a_n \rangle_{\mathbb{F}_q} \right)$

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- **Sum-rank weight** of $\mathbf{a} = (\mathbf{a}_1 \mid \dots \mid \mathbf{a}_\ell)$: $w_{SR}(\mathbf{a}) := \sum_{i=1}^{\ell} w_R(\mathbf{a}_i)$
where $\mathbf{a}_i \in \mathbb{F}_{q^m}^{n_i}$ and $n = \sum n_i$

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where $\mathbf{a}_i \in \mathbb{F}_{q^m}^{n_i}$ and $n = \sum n_i$
- The **minimum distance** of $\mathcal{C}$ is $d := \min_{c \in \mathcal{C}, c \neq 0} w(c)$.
- Notation $\mathcal{C} : [n, k, d]_{q^m}$
- Correction capacity of $\mathcal{C}$: $\lfloor \frac{d-1}{2} \rfloor$

Sum-Rank Decoding problem

Let $\mathbf{n} = (n_1, \dots, n_\ell) \in (\mathbb{N}^*)^\ell$, $\mathbf{r} = (r_1, \dots, r_\ell) \in \mathbb{N}^\ell$, $n = \sum_{i=1}^{\ell} n_i$ and $r = \sum_{i=1}^{\ell} r_i$.

Given $(\mathbf{G}, \mathbf{y}, \mathbf{r}) \in \mathbb{F}_{q^m}^{k \times n} \times \mathbb{F}_{q^m}^n \times \mathbb{N}^*$ with $\mathbf{G}$ of full rank, the $\text{SRD}(\mathbf{n}, k, \mathbf{r}, m)$ asks to

compute $\mathbf{e} = (\mathbf{e}_1 \mid \dots \mid \mathbf{e}_\ell) \in \mathbb{F}_{q^m}^n$ such that $\mathbf{y} = \mathbf{xG} + \mathbf{e}$, $\mathbf{e}_i \in \mathbb{F}_{q^m}^{n_i}$ and $w_R(\mathbf{e}_i) = r_i \ \forall i = 1, \dots, \ell$.

McEliece framework

- We need a code $\mathcal{C}$ with efficient decoding algorithm.
- Hide $\mathcal{C}$ to get a public code $\mathcal{C}_{pub}$.

Security:

- The indistinguishability of $\mathcal{C}_{pub}$ from a random code;
- The difficulty to decode with $\mathcal{C}_{pub}$.

1 Generalities

2 Linearized Reed-Solomon codes and their distinguishability

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Gabidulin codes

Let $\mathbf{u} = (u_1, \dots, u_n) \in (\mathbb{F}_{q^m}^*)^n$ such that $w_R(\mathbf{u}) = n$. $\theta : x \mapsto x^q$.

The **Gabidulin code** $\mathcal{G}_k(\mathbf{u})$ is a $[n, k, n - k + 1]_{q^m}$ with generator matrix

$$G_k^\theta(\mathbf{u}) = \begin{pmatrix} u_1 & \cdots & u_n \\ \theta(u_1) & \cdots & \theta(u_n) \\ \vdots & \vdots & \vdots \\ \theta^{k-1}(u_1) & \cdots & \theta^{k-1}(u_n) \end{pmatrix}$$

The capacity of correction of $\mathcal{G}_k(\mathbf{u})$ is $\lfloor \frac{n-k}{2} \rfloor$

Linearized Reed-Solomon codes

Let $a \in \mathbb{F}_{q^m}^*$, we define

$$D_k^\theta(a) = \text{Diagonal}(N_0^\theta(a), \dots, N_{k-1}^\theta(a))$$

a diagonal matrix where $N_i^\theta(a) = \prod_{j=0}^{i-1} \theta^j(a)$ and $N_0^\theta(a) = 1$.

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Definition (Martinez 2019)

Let $\ell, n, k, \eta \in \mathbb{N}^*$ such that $\ell \leq q - 1$, $\eta \leq m$, $n = \ell\eta$ and Let $k \leq n$.

Let $\gamma \in \mathbb{F}_{q^m}^*$ a primitive element and $i_1, \dots, i_\ell \in \{0, \dots, q - 2\}$ pairwise distinct with $i_1 = 0$.

Let $\mathbf{u}_i \in \mathbb{F}_{q^m}^\eta$ such that $w_R(\mathbf{u}_i) = \eta$, $\forall i = 1, \dots, \ell$.

A **linearized Reed-Solomon code** is a $[n, k, n - k + 1]_{q^m}$ code with generator matrix

$$\mathbf{G} = (G_k^\theta(\mathbf{u}_1) | D_k^\theta(\gamma^{i_2}) G_k^\theta(\mathbf{u}_2) | \dots | D_k^\theta(\gamma^{i_\ell}) G_k^\theta(\mathbf{u}_\ell)).$$

Codes	$[n, k]_{q^m}$ Gabidulin code	$[n, k]_{q^m}$ Linearized Reed–Solomon code
Dual	Gabidulin code	Linearized Reed–Solomon code
Length	$n \leq m$	$n \leq (q - 1)m$
Decoding algorithm	Efficient decoding in rank metric	Efficient decoding in sum-rank metric
Capacity of correction	$\frac{n - k}{2}$	$\frac{n - k}{2}$
Distinguisher	Overbeck distinguisher (Polynomial time)	Overbeck-type distinguisher (Exponential time)

Overbeck distinguisher

Let $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{F}_{q^m}^n$. For any $i \in \mathbb{N}$, we define $\theta^i(\mathbf{x}) = (\theta^i(x_1), \dots, \theta^i(x_n))$.

Let $\ell \in \mathbb{N}$ and $\mathcal{C}$ a $[n, k]_{q^m}$ code. We define

$$\Gamma_\ell^\theta(\mathcal{C}) = \mathcal{C} + \theta(\mathcal{C}) + \dots + \theta^\ell(\mathcal{C}).$$

- Let $\mathcal{G}$ a $[n, k]_{q^m}$ Gabidulin code, if $\ell \leq n - k$ then $\dim(\Gamma_\ell^\theta(\mathcal{G})) = \ell + k$.
- Let $\mathcal{C}_{\text{rand}}$ a $[n, k]_{q^m}$ random code, $\dim(\Gamma_\ell^\theta(\mathcal{C}_{\text{rand}})) = \min(n, (\ell + 1)k)$.

Linearized Reed-Solomon codes distinguisher

Consider $\mathcal{C}$ a $[2\eta, k]$ linearized Reed-Solomon code with generator matrix

$$G = (G_k^\theta(\mathbf{u}_1) | D_k^\theta(\gamma) G_k^\theta(\mathbf{u}_2)) = \begin{pmatrix} u_{11} & \cdots & u_{1\eta} & N_0^\theta(\gamma) u_{21} & \cdots & N_0^\theta(\gamma) u_{2\eta} \\ \theta(u_{11}) & \cdots & \theta(u_{1\eta}) & N_1^\theta(\gamma) \theta(u_{21}) & \cdots & N_1^\theta(\gamma) \theta(u_{2\eta}) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \theta^{k-1}(u_{11}) & \cdots & \theta^{k-1}(u_{1\eta}) & N_{k-1}^\theta(\gamma) \theta^{k-1}(u_{21}) & \cdots & N_{k-1}^\theta(\gamma) \theta^{k-1}(u_{2\eta}) \end{pmatrix}.$$

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- We can not use Overbeck distinguisher.

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- We can not use Overbeck distinguisher.
- In this work, we propose a polynomial distinguisher by combining Overbeck and duality.

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New encryption scheme

Key Generation

- $\mathbf{u}_1, \mathbf{u}_2 \xleftarrow{\$} \mathbb{F}_{q^m}^\eta$, $w_R(\mathbf{u}_1) = w_R(\mathbf{u}_2) = \eta = \frac{n}{2}$
 - $\gamma \xleftarrow{\$} \mathbb{F}_{q^m}^*$ primitive element
 - $\mathbf{G} = (G_k^\theta(\mathbf{u}_1) \mid D_k^\theta(\gamma)G_k^\theta(\mathbf{u}_2))$
 - $\mathcal{V}_1, \mathcal{V}_2 \xleftarrow{\$} \lambda$ -dimensional subspaces of $\mathbb{F}_{q^m}$
 - $\mathbf{P}_i \xleftarrow{\$} GL_\eta(\mathbb{F}_{q^m}) \cap \mathcal{M}_\eta(\mathcal{V}_i)$
 - $\mathbf{P} = \begin{pmatrix} \mathbf{P}_1 & 0 \\ 0 & \mathbf{P}_2 \end{pmatrix}$
 - $\mathbf{S} \xleftarrow{\$} GL_k(\mathbb{F}_{q^m})$
 - Compute $\mathbf{G}_{\text{pub}} = \mathbf{S}\mathbf{G}\mathbf{P}^{-1}$
- Public key : $(\mathbf{G}_{\text{pub}}, \frac{n-k}{2\lambda})$
- Secret key : $(\gamma, \mathbf{u}, \mathbf{v}, \mathbf{S}, \mathbf{P})$

Encryption of $\mathbf{m} \in \mathbb{F}_{q^m}^k$

- $\mathbf{e} = (\mathbf{e}_1 \mid \mathbf{e}_2) \xleftarrow{\$} \mathbb{F}_{q^m}^n$ with $w_{\text{SR}}(\mathbf{e}) \leq \frac{n-k}{2\lambda}$
- Compute $\mathbf{y} = \mathbf{m}\mathbf{G}_{\text{pub}} + \mathbf{e}$

Decryption of $\mathbf{y}$

- Compute $\mathbf{y}\mathbf{P} = \mathbf{m}\mathbf{S}\mathbf{G} + \mathbf{e}\mathbf{P}$
- Decode $\mathbf{y}\mathbf{P}$ and use $\mathbf{S}^{-1}$ to recover $\mathbf{m}$.

$$\begin{aligned} w_{\text{SR}}(\mathbf{e}\mathbf{P}) &= w_R(\mathbf{e}_1\mathbf{P}_1) + w_R(\mathbf{e}_2\mathbf{P}_2) \\ w_{\text{SR}}(\mathbf{e}\mathbf{P}) &\leq \lambda(w_R(\mathbf{e}_1) + w_R(\mathbf{e}_2)). \end{aligned}$$

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Distinguisher and structural attacks

Let $\mathcal{C}_{\text{pub}}$ the public code generated by $\mathbf{G}_{\text{pub}}$.

Case $\ell = 1$: We have

$$\mathbf{G}_{\text{pub}} = \mathbf{S} G_k^\theta(\mathbf{u}_1) \mathbf{P}_1^{-1}.$$

In this case the scheme corresponds to the Loidreau's scheme [Loidreau 2017].

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- Coggia-Couvreur distinguisher: $\mathcal{C}_{\text{pub}}$ is different from a random code if

$$\lambda \dim \mathcal{C}_{\text{pub}}^\perp + \lambda < \min(n, (1 + \lambda) \dim \mathcal{C}_{\text{pub}}^\perp).$$

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- Structural attack:
If the distinguisher is applicable and $\lambda = 2$ ([Coggia, Couvreur 2020]), or $\lambda = 3$ ([Ghatak, 2022]).

Case $\ell \neq 1$:

Consider a $[\ell\eta, k]$ code $\mathcal{C}_{\text{pub}}$ with generator matrix

$$\mathbf{G}_{\text{pub}} = \mathbf{S}(G_k^\theta(\mathbf{u}_1)|D_k^\theta(\gamma^{i_2})G_k^\theta(\mathbf{u}_2)|\cdots|D_k^\theta(\gamma^{i_\ell})G_k^\theta(\mathbf{u}_\ell))\mathbf{P}^{-1}.$$

Let $b = \lfloor \frac{k}{\eta} \rfloor + 1$

The public code $\mathcal{C}_{\text{pub}}$ is distinguished from a random code if

$$\lambda(b\eta - k) + \lambda < \min(\eta, (1 + \lambda)(b\eta - k))$$

Sum-Rank Decoding problem

To solve problem $\text{SRD}(\mathbf{n}, k, \mathbf{r}, m)$, we can use the following attack.

- Combinatorial attack [Gaborit et al, 2024]

$$\mathcal{O} \left((n - k)^\omega m^\omega q^{-m + \sum_{i=1}^{\ell} r_i(m - t_i)} \right)$$

where ω is the linear algebra constant, $r_i \leq t_i \leq m$ and $\sum_{i=1}^{\ell} n_i t_i \leq m(n - k)$.

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- Algebraic attack [Huang et al, 2023]

$$\mathcal{O} \left(q^{\sum_{i=1}^{\ell} a_i r_i} m^{\binom{n - k - 1}{r}} \left(\prod_{i=1}^{\ell} \binom{n_i - a_i}{r_i} \right)^{\omega - 1} \right)$$

where $(a_1, \dots, a_\ell) \in \mathbb{N}^\ell$ such that $m^{\binom{n - k - 1}{r}} \geq \prod_{i=1}^{\ell} \binom{n_i - a_i}{r_i} - 1$.

Proposition of parameters

We propose the following parameters.

m	n	k	η	t	λ	q	Sec. Target	pk-size	ct-size
25	75	27	25	12	2	4	136	7.91 kB	0.29 kB
32	90	30	30	15	2	4	200	14.06 kB	0.47 kB
36	108	36	36	18	2	4	310	24.04 kB	0.63 kB

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Comparison with other schemes

Scheme	128 bits	192 bits
Our scheme	8.2 kB	14.53 kB
RQC	5.48 kB	8.54 kB
LowMS	5.76 kB	14.97 kB
HQC	6.73 kB	13.56 kB
Loidreau scheme	35.8 kB	36 kB
Classic McEliece	261.2 kB	624.3 kB

Conclusion

We propose a new McEliece-type encryption scheme in sum-rank metric.

We obtain the small size of the key.

We can use this to construct a KEM.

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Thank you for your attention.